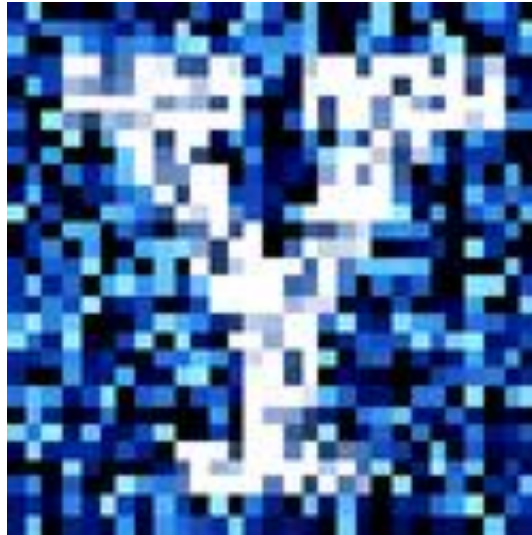


YData: Introduction to Data Science



Class 06: Array computations

Overview

Brief review of statistics and visualizations

- Finishing up z-score and correlations

NumPy arrays

- Creating arrays
- Array computations
- Boolean masking



Announcement

Sign up for practice session!

Homework 2 is due on Gradescope on **Sunday Feb. 1st at 11pm**

- **Be sure to mark each question on Gradescope!**

Homework 3 has been posted!

It is due on Gradescope on **Sunday Feb. 8th at 11pm**

- **Be sure to mark each question on Gradescope!**

Quick review: one quantitative variable data

Basics statistics and plots:

`plt.hist(data)`

`statistics.mean(data)`

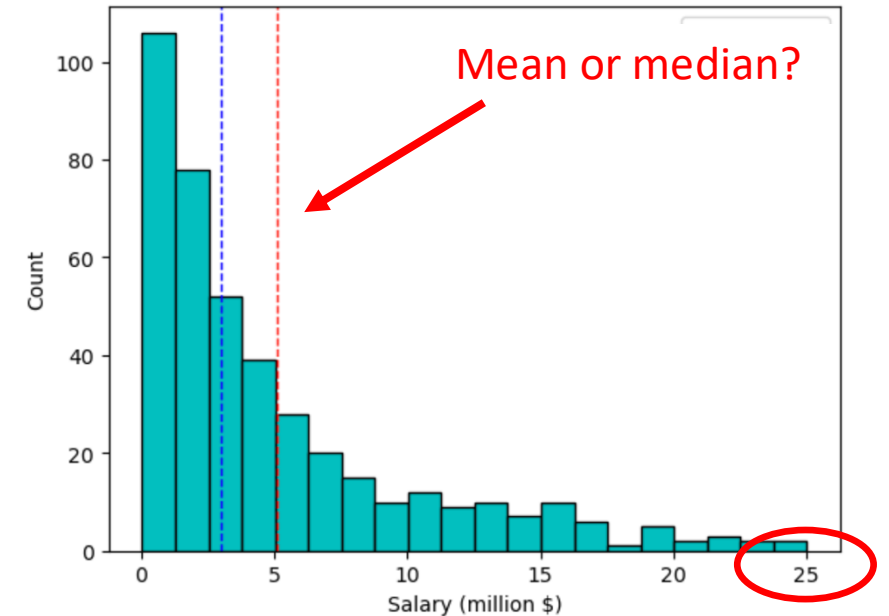
`statistics.median(data)`

$$= \frac{1}{n} \sum_{i=1}^n x_i$$

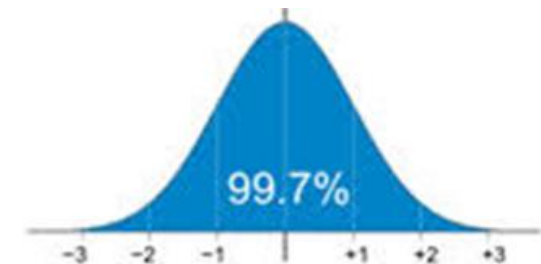
Quantitative data spread

`statistics.stdev(data)`

$$s = \sqrt{\frac{1}{(n-1)} \sum_{i=1}^n (x_i - \bar{x})^2}$$



Outliers



Z-scores

Standardized units

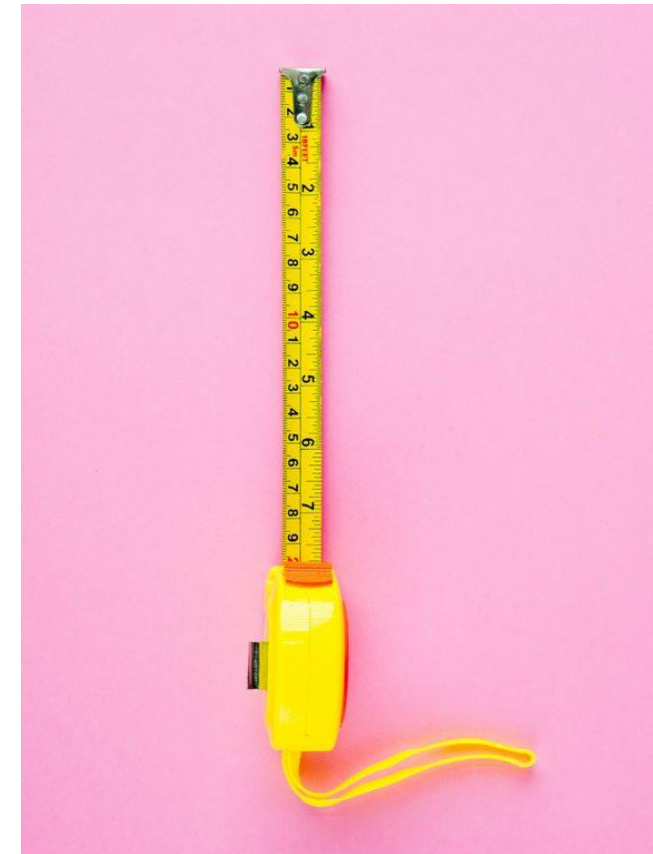
Item in the world are often measured on very different scales

How can we create a standard scale to quantify unusual/large/impressive values?

Z-scores measure how many SDs a value is from average:

$$\text{z-score}(x_i) = \frac{x_i - \bar{x}}{s}$$

- Negative z: value below average
- Positive z: value above average
- $z = 0$: value equal to average



Which Accomplishment is most impressive?

LeBron James is a basketball player who had the following statistics in 2011:

- Field goal percentage (FGPct) = 0.510
- Points scored = 2111
- Assists = 554
- Steals = 124

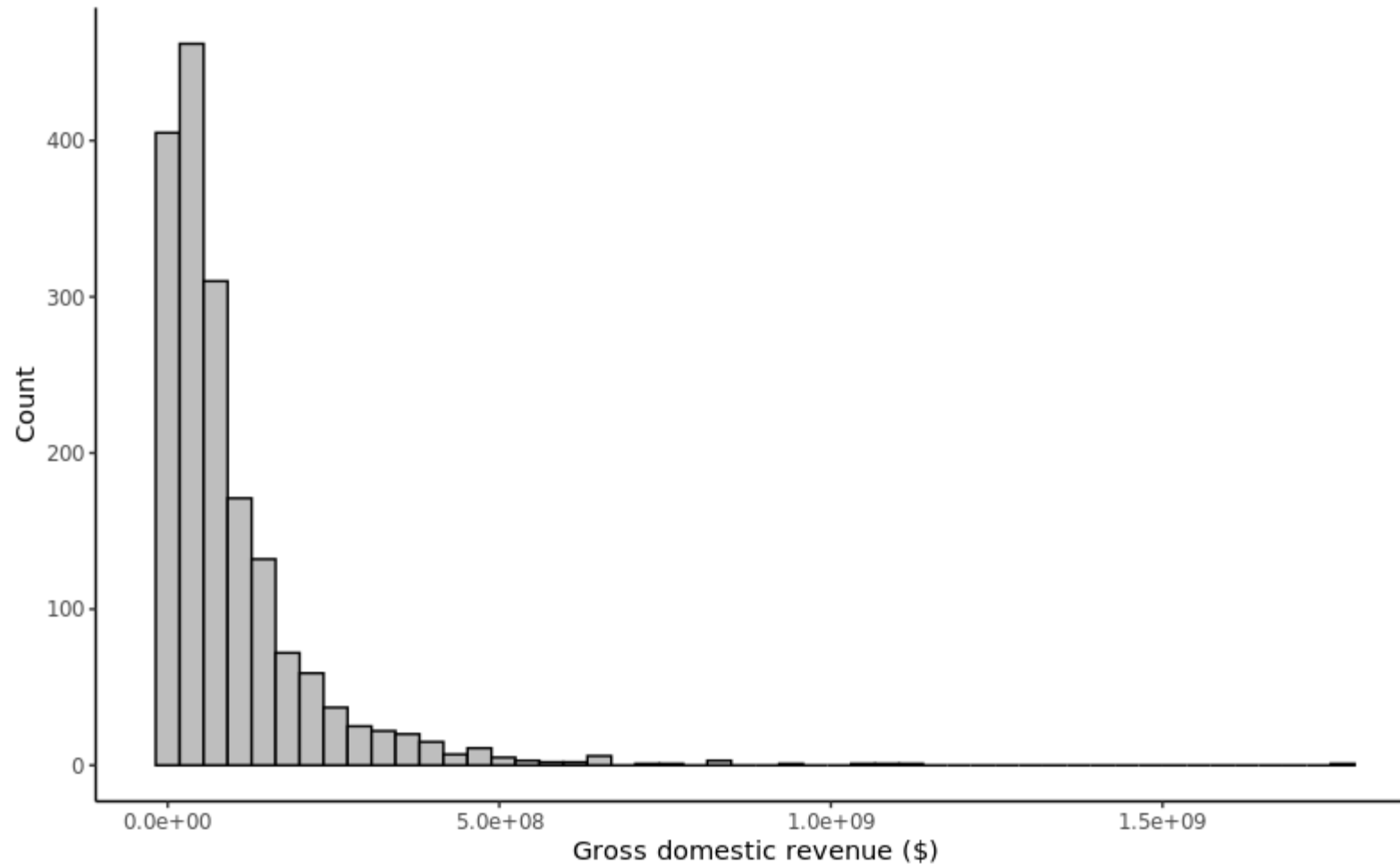


The summary statistics of the NBA in 2011 are given below

	Mean	Standard Deviation
$\text{z-score}(x_i) = \frac{x_i - \bar{x}}{s}$		
FGPct	0.464	0.053
Points	994	414
Assists	220	170
Steals	68.2	31.5

Question: Relative to his peers, which statistic is most and least impressive?

What is star wars' z-score?



\$1,771,682,790

Let's try it in Python!

Relationships between two
quantitative variables

Do movies with larger budgets make more money?

Q: How could we visualize the data to see if this is true?

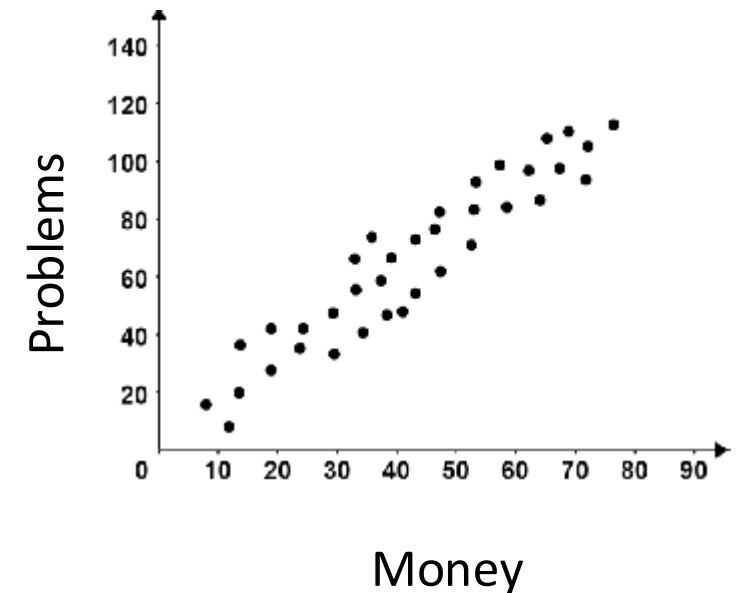


Visualizing two quantitative variables: scatterplots

A **scatterplot** graphs the relationship between two variables

- Each axis represents the value of one variables
- Each point the plot shows the value for the two variables for a single data case

If there is an explanatory and response variable, then the explanatory variable is put on the x-axis and the response variable is put on the y-axis



```
plt.plot(x, y, '.')
```

Do movies with larger budgets make more money?

Q: If we want to create a scatter plot to address whether movies with larger budgets make more money, what variables should we use in our plot?

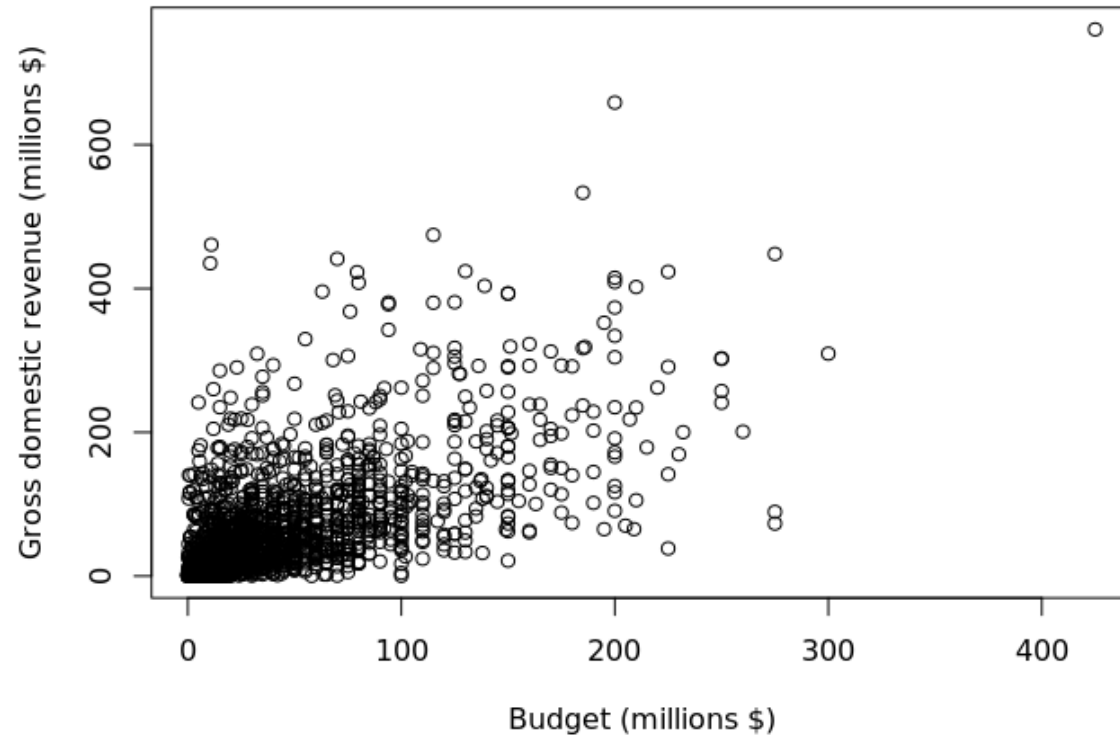
A:

- budget_2013
- domgross_2013



Relationship movie money spent and made

Bechel movies relationship between buget and revenue



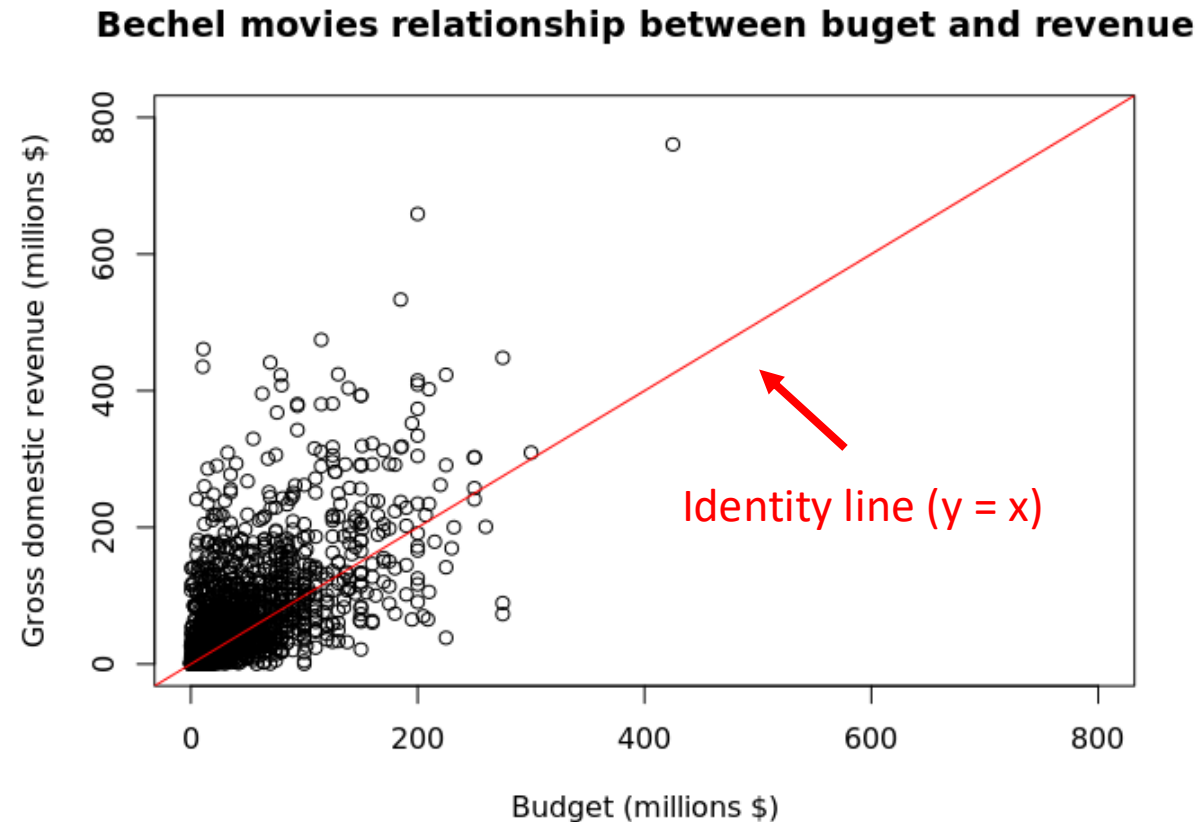
Do movies with larger budgets make more money?

Do most movies make money?

- How could we create a more informative scatter plot of this data?

Matplotlib: `plt.plot(x, y)`

Relationship movie money spent and made



Do movies with larger budgets make more money?

Do most movies make money?

- How could we create a more informative scatter plot of this data?

Matplotlib: `plt.plot(x, y)`

Questions when looking at scatterplots

Do the points show a clear trend?

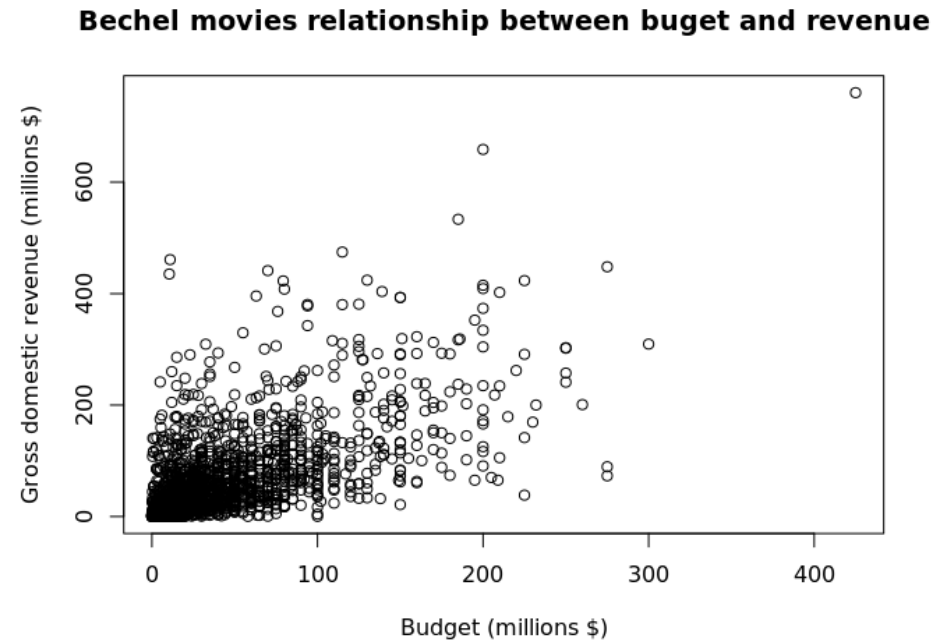
Does it go upward or downward?

How much scatter around the trend?

Does the trend seem be linear (follow a line) or is it curved?

Are there any outlier points?

Budget and revenue



Positive, negative, no correlation

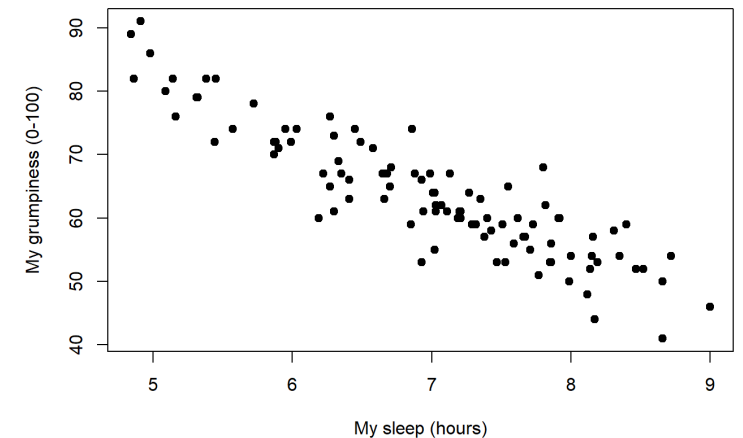
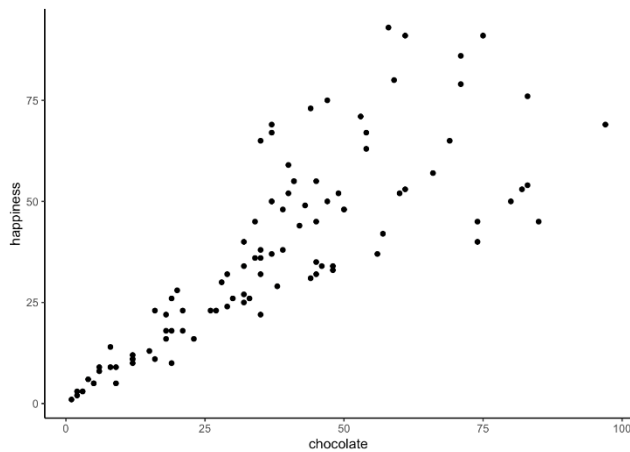
Do the points show a clear trend?

Does it go upward or downward?

How much scatter around the trend?

Does the trend seem be linear (follow a line) or is it curved?

Are there any outlier points?



The correlation coefficient

The **correlation** is measure of the strength and direction of a linear association between two variables

$$r = \frac{1}{(n-1)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

```
statistics.correlation(x, y)
```

Properties of the correlation

Correlation is always between -1 and 1: $-1 \leq r \leq 1$

The sign of r indicates the direction of the association

Values close to ± 1 show strong linear relationships, values close to 0 show no linear relationship

Correlation is symmetric: $r = \text{cor}(x, y) = \text{cor}(y, x)$

$$r = \frac{1}{(n-1)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

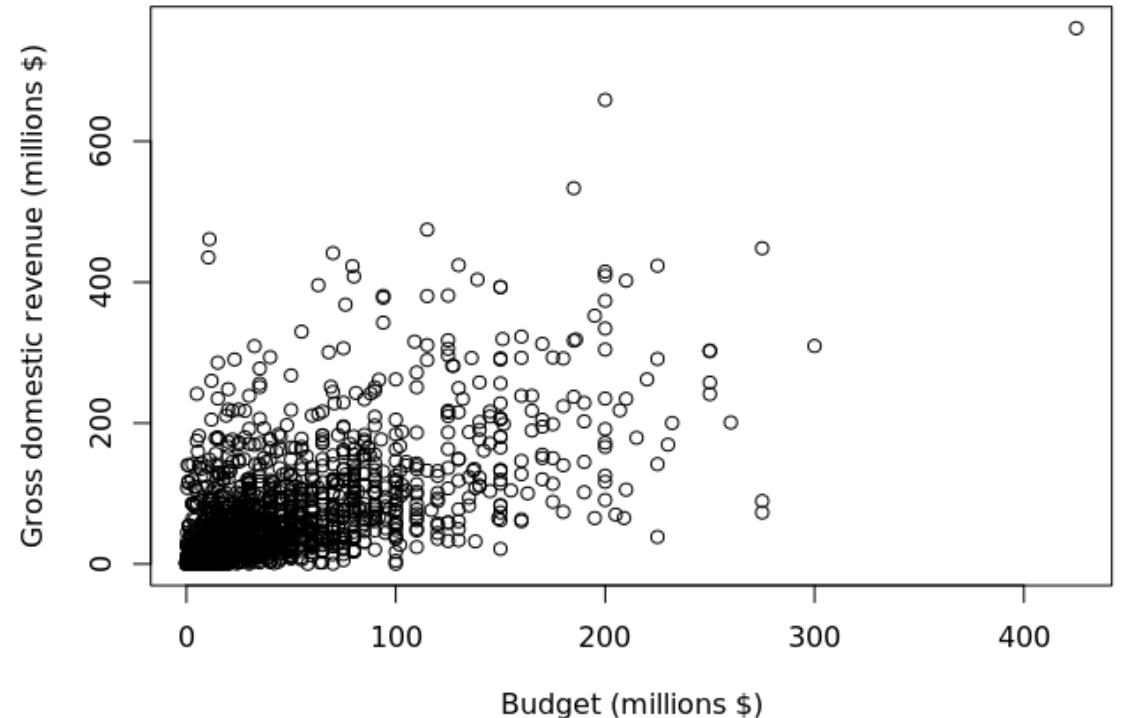
Movie budget and revenue correlation?

The **correlation** is measure of the strength and direction of a linear association between two variables

$r = ?$

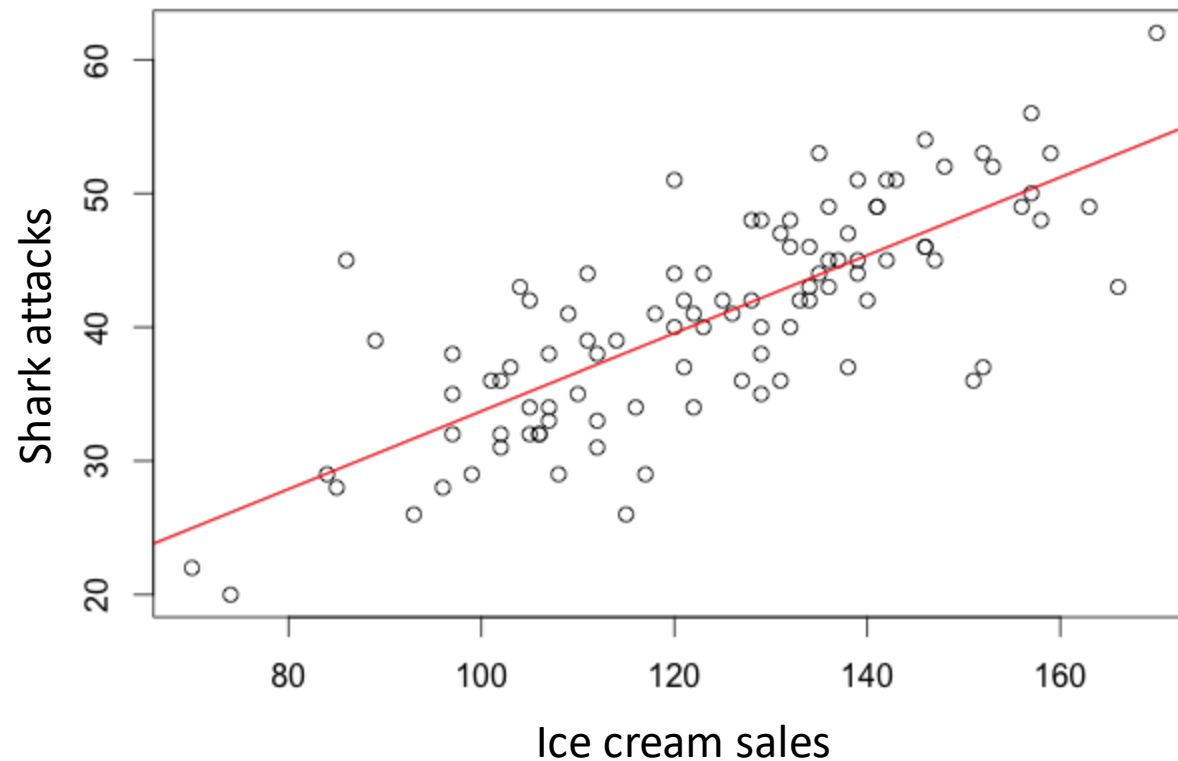
Let's calculate the correlation in Python!

Bechel movies relationship between buget and revenue



Correlation caution #1

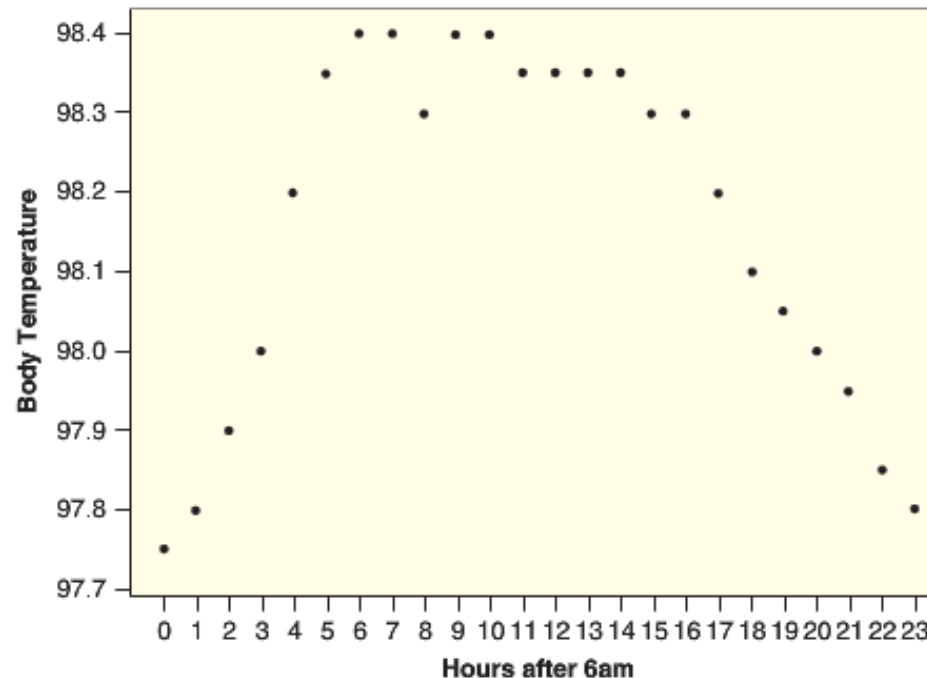
A strong positive or negative correlation does not (necessarily) imply a cause and effect relationship between two variables



Correlation caution #2

A correlation near zero does not (necessarily) mean that two variables are not associated. Correlation only measures the strength of a linear relationship.

Example: Body temperature as a function of time of the day



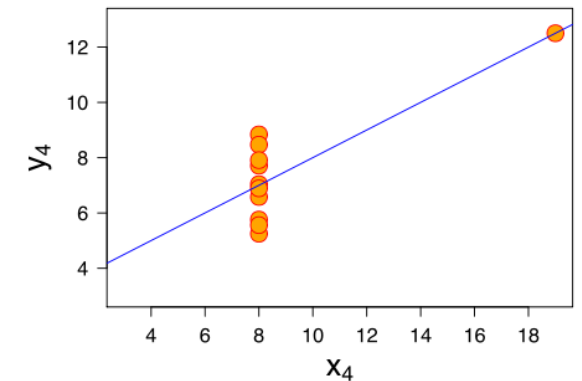
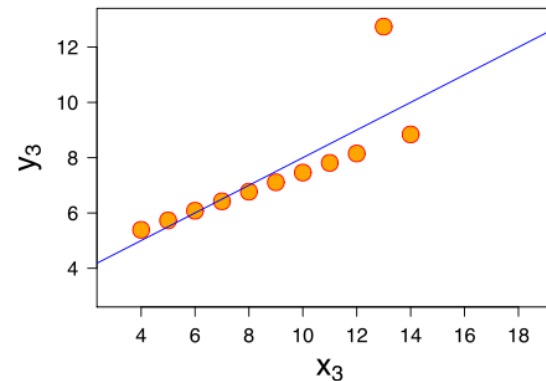
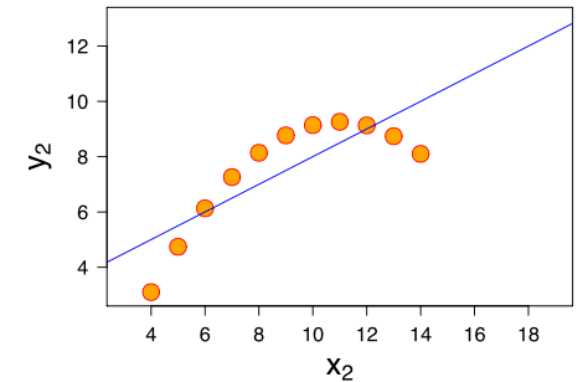
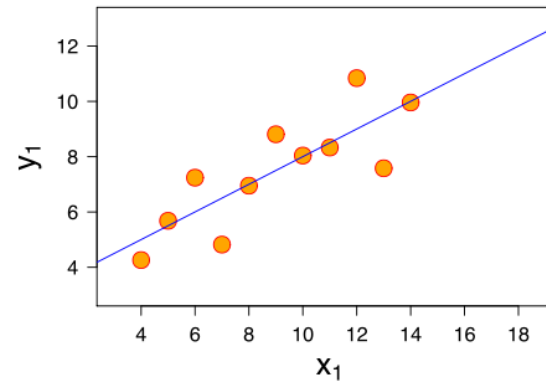
$r = -0.08$

Correlation caution #3

Correlation can be heavily influenced by outliers. Always plot your data!

Example:
“Anscombe’s quartet”

$r = 0.81$ for all plots



Additional exercises: NBA salaries

Let's practice analyzing data by explore salaries of NBA players

- (from the 2022-2023 season)



Player Name	Salary	Position	Team	PTS
Stephen Curry	48070014	PG	GSW	29.4
John Wall	47345760	PG	LAC	11.4
Russell Westbrook	47080179	PG	LAL/LAC	15.9
LeBron James	44474988	PF	LAL	28.9
Kevin Durant	44119845	PF	BRK/PHO	29.1
Bradley Beal	43279250	SG	WAS	23.2
Kawhi Leonard	42492492	SF	LAC	23.8
Paul George	42492492	SF	LAC	23.8
Giannis Antetokounmpo	42492492	PF	MIL	31.1
Damian Lillard	42492492	PG	POR	32.2

Array computations

Arrays

Often, we are processing data that is all of the same type

- For example, we might want to do processing on a data set of numbers
 - e.g., if we were analyzing salary data

When we have data that is all of the same type, there are more efficient ways to process data than using a list

- i.e., methods that are faster and take up less memory

In Python, the *NumPy package* offers ways to store and process data that is all of the same type using a data structure called a ***ndarray***

There are also functions that operate on ndarrays that can do computations very efficiently



ndarrays

We can import the NumPy package using: `import numpy as np`

We can then create an array by passing a list to the `np.array()` function

- `my_array = np.array([1, 2, 3])`

We can get elements of an array using similar syntax as using a list

- `my_array[1]` `# what does this return`

ndarrays have properties that tell us the type and size

- `my_array.dtype` `# get the type of elements stored in the array`
- `my_array.shape` `# get the dimension of the array`
- `my_array.astype('str')` `# convert the numbers to strings`
- `sequential_nums = np.arange(1, 10)` `# creates numbers 1 to 9`

Let's explore this in Jupyter!

NumPy functions on numerical arrays

The NumPy package has a number of functions that operate very efficiently on numerical ndarrays

- `np.sum()`, `np.max()`, `np.min()`
- `np.mean()`, `np.median()`
- `np.diff()` # takes the difference between elements
- `np.cumsum()` # cumulative sum
- `np.prod()` # calculates the product of all elements in an array

$$\prod_{i=1}^n x_i$$

There are also "broadcast" functions that operate on all elements in an array

- `my_array = np.array([12, 4, 6, 3, 4, 3, 7, 4])`
- `my_array * 2`
- `my_array2 = np.array([10, 9, 2, 8, 9, 3, 8, 5])`
- `my_array - my_array2`

FRED: Federal Reserve Economic Data

[Federal Reserve Economic Data \(FRED\)](#) is a database maintained by the Research division of the Federal Reserve Bank of St. Louis that has more than 816,000 economic time series from various sources.



They cover:

- E.g., Consumer price indexes
- Employment and population
- Gross domestic product
- Producer price indexes
- Etc.

We can read this data into Python using `pandas_datareader`

```
from pandas_datareader.fred import FredReader
```

```
FredReader("GASREGW").read()
```

Let's explore numpy functions using the [price of gas](#) from FRED!

Boolean arrays

It is often to compare all values in an ndarray to a particular value

- `my_array = np.array([12, 4, 6, 3, 4, 3, 7, 4])`
- `my_array < 5` # any guesses what this will return
 - `array([False, True, False, True, True, True, False, True])`

This can be useful for calculating proportions

- `True == 1` and `False == 0`
- Taking the sum of a Boolean array gives the total number of `True` values
- The number of `True` 's divided by the length is the proportion
 - Or we can use the `np.mean()` function

Categorical Variable

PLAYER	POSITION	TEAM	SALARY
str	str	str	f64
"Paul Millsap"	"PF"	"Atlanta Hawks"	18.671659
"Al Horford"	"C"	"Atlanta Hawks"	12.0
"Tiago Splitter..."	"C"	"Atlanta Hawks"	9.75625
"Jeff Teague"	"PG"	"Atlanta Hawks"	8.0
"Kyle Korver"	"SG"	"Atlanta Hawks"	5.746479

Proportion centers =

number of centers

total number

Let's explore this in Jupyter!

Boolean masking

We can also use Boolean arrays to return values in another array

- This is called "Boolean masking", "Boolean subsetting" or "Boolean indexing"

```
my_array = np.array([12, 4, 6, 3])  
boolean_mask = np.array([False, True, False, True])  
  
smaller_array = my_array[boolean_mask]
```

This can be useful for calculating statistics on data that meet particular criteria:

- `np.mean(my_array[my_array < 5])` # what does this do?

Boolean masking

Suppose you wanted to get the average movie revenue for movies that passed the Bechdel test

- [domgross_2013](#): Movie revenue
- [bechdel](#): whether a movie passed the Bechdel test

Can you do it?



Let's explore this in Jupyter!